

Explaining Data Mixing Scaling Laws

Harvard EconCS Seminar

Rui Dai, and Shuran Zheng

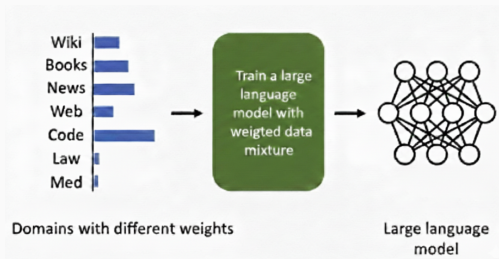
Feb. 2026

The Critical Role of Data Mixing

- Large foundation models are trained on data from multiple domains (e.g., GitHub, books, Wikipedia).

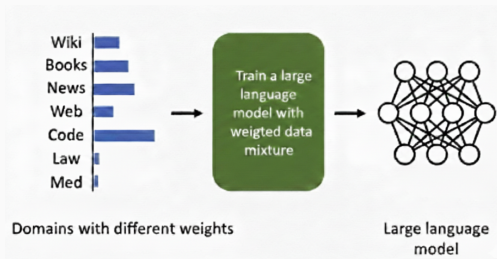
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- **The Data Mixture:** The proportion of each domain used is a critical factor in determining model performance.



- **Key question:** How do we find the best domain weights to optimize the model's performance?

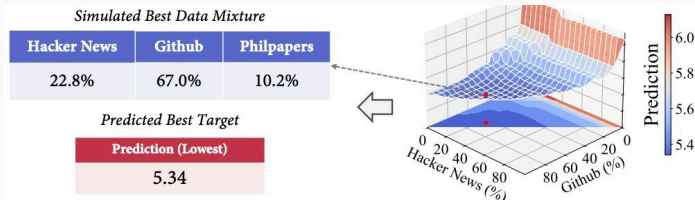
Recent Research: Empirical Data Mixing Laws

Data mixing laws: Various functional forms to model the relationship between weights and domain losses (Ye et al., 2024, Ge et al., 2024, Shukor et al., 2025)

$$\text{Loss} = f(\text{domain weights})$$

$$\text{Optimal weights} = \arg \min f$$

- Sample random weights $\rightarrow$ train **small proxy models** $\rightarrow$ fit their losses with hand-picked functions.



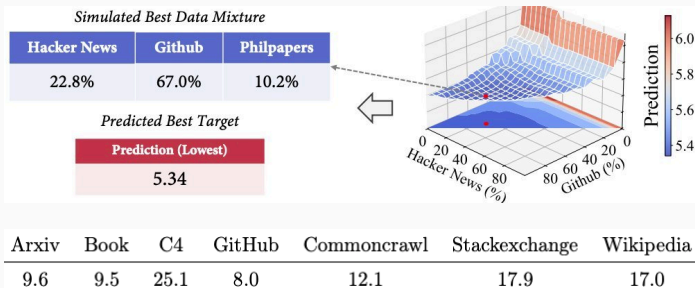
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Empirical Data Mixing Laws

- **State of the art:**

- Given a fixed model size N , a fixed data size D
- L_i : loss on domain i
- $h_i \in (0, 1)$: domain i 's weight

Reference	Functional Form ($f_i(h, N, D)$)
(Shukor et al., 2025)	$L_i \approx E_i + \left(\sum_{j=1}^K C_{ij} h_j^{\gamma_{ij}} \right)^{-1}$
(Ye et al., 2024)	$L_i \approx c_i + k_i \exp \left(\sum_{j=1}^K t_{ij} h_j \right)$

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- **Domain interaction:**

- L_i depends not only on h_i , but also the weights of other domains h_{-i}
- This correlation does not exhibit a simple functional form.

Empirical Data Mixing Laws: Limitations

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Limitation: There is **no theoretical understanding** of the underlying mechanics driving domain interaction.

- Do they generalize to larger scales?
- Do they generalize to different datasets?
- Map the predicted domain loss to downstream task performance?

A theoretical framework that extends theories for standard neural scaling laws

Two-level framework

Two key factors

Extended Quantization Model

Capacity competition

**Extended Projected Linear
Regression Model**

Noise decided by data amount

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- Skill learning ([Michaud et al. 2023](#))

Extended Projected Linear Regression Model

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- Projected Linear Regression trained with one-pass SGD ([Bordelon et al., 2024](#), [Lin et al., 2024](#))

A theoretical framework that extends theories for standard neural scaling laws

Two-level framework

Two key factors

Extended Quantization Model

- Skill learning ([Michaud et al. 2023](#))

Capacity competition

- Domains competing for the finite model capacity
- Introduces a non-trivial domain interaction

Extended Projected Linear Regression Model

- Projected Linear Regression trained with one-pass SGD ([Bordelon et al., 2024](#), [Lin et al., 2024](#))

Noise decided by data amount

- Domain losses depend on the data amount
- Shifts the optimal mixture weights toward more difficult domains

Experimentally, our model achieves

- **Superior Fitting Accuracy:** Our models fit the observed loss landscape with lowest Mean Relative Error (MRE)
- **Optimal Mixture Prediction:** Our framework identifies optimal training mixtures that yield the lowest validation loss.
- **Parameter Efficiency:** Our models utilize significantly fewer free parameters compared to leading empirical laws.

Problem Description

Extended Quantization Model: Capacity Competition

Extended Linear Model: Noise Driven by Data Amount

Future Work

Problem Description

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Key Challenge

Approximate $L_i(h \mid \cdot)$!

Key Properties of Loss Landscape and Optimal Mixture

- Next: A theoretically grounded framework that predicts $L_i(h \mid \cdot)$

Key Properties of Loss Landscape and Optimal Mixture

- **Next: A theoretically grounded framework that predicts $L_i(h \mid \cdot)$**
- Two key properties of the loss landscape and optimal mixture observed in empirical work.
 - **Domain interaction:**

$$L_i(h) = \frac{1}{\sum_{j=1}^K C_{ij} h_j^{\gamma_{ij}}} + E_i$$

- **Non-uniform optimal mixture:**

Arxiv	Book	C4	GitHub	Commoncrawl	Stackexchange	Wikipedia
9.6	9.5	25.1	8.0	12.1	17.9	17.0

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- **Mechanism:**
 - The model has a finite capacity N (total "quanta" of skills it can learn).
 - It allocates this capacity across different domains to minimize the total loss.
- **Strength:** Offering an explanation for the domain interaction

Single-Domain Foundation

Quantization Model (Michaud et al., 2023): Each domain $\mathcal{D}_i$ consists of a continuous space of skills indexed by $k_i \in [1, \infty)$ following a **power-law density**:

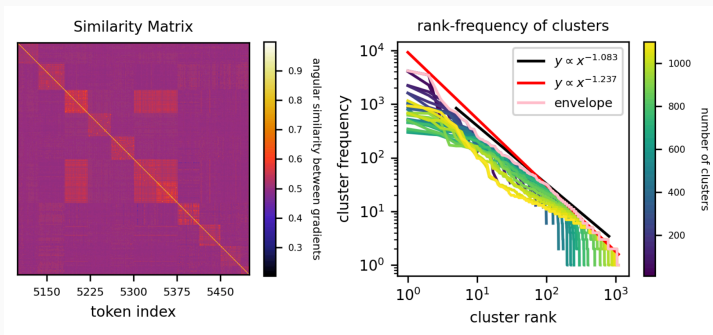
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- Experiments on the TinyStories dataset from (Michaud et al., 2023):



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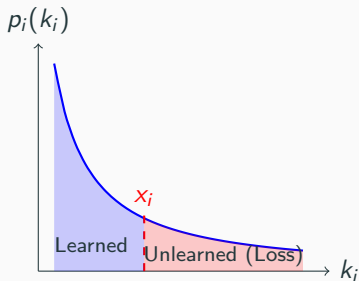
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- **Learning Process:** A model with allocated capacity x_i learns all high-frequency skills up to threshold x_i ($k_i \leq x_i$).
- **Domain Loss (L_i)** = The probability mass of the unlearned tail ($k_i > x_i$) multiplied by a **domain-specific constant** c_i :

$$L_i = c_i \int_{x_i}^{\infty} (\alpha_i - 1) k_i^{-\alpha_i} dk_i = c_i x_i^{-b_i}$$

where $b_i = \alpha_i - 1$.

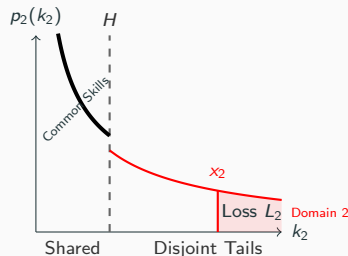
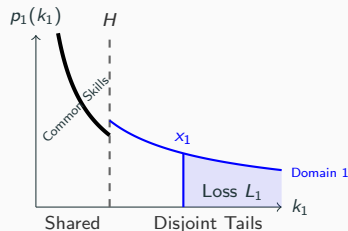


Structural Assumption: Shared Head, Disjoint Tail

We assume a "Shared Head, Disjoint Tail" data structure:

- **Shared Head ($k \leq H$):** Domains overlap on fundamental, high-probability skills (e.g., basic logic, grammar).
- **Disjoint Tail ($k > H$):** Rare, specialized skills are unique to each domain and independent of others.

Implication: The model easily learns the shared head. The core challenge is **allocating the remaining capacity ($N - H$) among the disjoint specialized tails.**



Training as Capacity Allocation

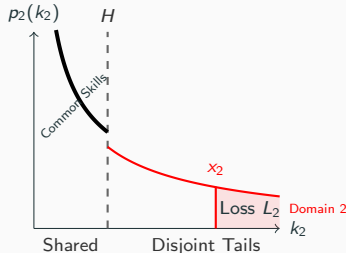
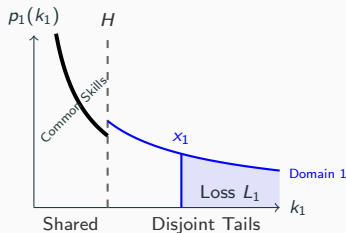
When trained on a mixture $h = (h_1, \dots, h_K)$, the model implicitly solves this constrained optimization problem:

Capacity Allocation Problem

$$\begin{aligned} \min_{\{x_i\}} \quad & \sum_{i=1}^K h_i L_i = \sum_{i=1}^K h_i c_i x_i^{-b_i} \\ \text{s.t.} \quad & \sum_{i=1}^K (x_i - H) \leq N - H \end{aligned}$$

Domain Loss: $L_i(h) = c_i(x_i^*(h))^{-b_i}$

Competition: The constraint couples all domains, introducing a non-trivial correlation.



Solution and Domain Interaction

The optimal threshold $x_i^*(h)$ does not have a closed-form solution. Applying Lagrange multipliers yields:

- **Stationarity Condition:** The gradient of the Lagrangian $\mathcal{L}$ with respect to x_i yields:

$$h_i c_i b_i (x_i^*)^{-(b_i+1)} = \lambda \implies x_i^*(h) = \left(\frac{h_i c_i b_i}{\lambda} \right)^{\frac{1}{b_i+1}}$$

- **Determining the Multiplier λ :** The shared Lagrange multiplier λ is implicitly defined by the capacity constraint:

$$\sum_{k=1}^K \left(\frac{h_k c_k b_k}{\lambda} \right)^{\frac{1}{b_k+1}} = N$$

Key Insight: Global Coupling via λ

Domains are coupled through the shadow price λ .

Empirical Validation

Fitting Accuracy (trained models from (Liu et al. 2024))

- **Models:** 1B-parameter models trained on 25B tokens.
- **Domains:** 17 domains from the Pile dataset. 64 mixtures.
- **Metric:** Mean Relative Error $\left| \frac{\hat{L}_i - L_i}{L_i} \right|$ and Mean Absolute Error $|\hat{L}_i - L_i|$ on held-out mixtures.

Method	MRE (%) ↓	MAE ↓	# Param
<i>Empirical Baselines</i>			
Exponential (Ye et al.)	6.990	0.059	$K(K + 2)$
RegMix (Liu et al.)	6.480	0.136	K^2
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<i>Theoretical Models (Ours)</i>			
Extended Quantization	2.064	0.051	$3K$

Table 1: Fitting accuracy on 1B-parameter models ($K = 17$ domains).

Limitation: Optimal Mixture Failure

We formulate the search for the optimal training mixture h^* as a **bi-level optimization problem**:

- **Outer Level (Target Objective):** Find h to minimize loss on the *target distribution* w :

$$h^* = \arg \min_h \sum_{i=1}^K w_i L_i(x^*(h)) \quad (1)$$

- **Inner Level (Training Dynamics):** The model allocates capacity $x^*(h)$ to minimize loss on the *training mixture* h :

$$x^*(h) = \arg \min_x \sum_{i=1}^K h_i L_i(x) \quad \text{s.t.} \quad \sum x_i = N \quad (2)$$

The Limitation: Trivial Solution

- The optimal strategy is simply to set training weights equal to target weights: $h^* = w$.
- **Contradiction:** Empirical studies show h^* deviates from w .

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- Training = static capacity allocation. It ignores the training dynamics.
- Real-world models are trained via **stochastic optimizers** (e.g., Adam, SGD), which introduce gradient noise.

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- **Next Step:**

- We follow (Bordelon et al., 2024, Lin et al., 2024) and model the training process of LLM via a **Projected Linear Regression Model** trained with **one-pass SGD**.

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- **Key Idea:**
 - Model the training process of LLM via a **Projected Linear Regression Model** trained with **one-pass SGD**
 - The SGD training process **implicitly solves the capacity allocation problem**
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 - The SGD training process **implicitly solves the capacity allocation problem**
 - It introduces an **additional noise term** that depends on data amount
- **Strength:**
 - Explains why the optimal mixture h^* deviates from the target weights w .

Problem Formulation

Consider a d -dimensional linear regression problem with large d :

$$y = \langle \theta^*, x \rangle + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2) \quad (3)$$

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$$\tilde{x} = Sx \text{ with } S \in \mathbb{R}^{N \times d}.$$

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$$\tilde{x} = Sx \text{ with } S \in \mathbb{R}^{N \times d}.$$

Training dynamics: The model learns $\hat{w} \in \mathbb{R}^N$ via **one-pass SGD**, minimizing the squared error on the projected features $\tilde{x} = Sx$:

$$\hat{w} = \arg \min_{w \in \mathbb{R}^N} \frac{1}{D} \sum_{i=1}^D (y_i - w^\top Sx_i)^2$$

Problem Formulation: Spectral Structure

We analyze the covariance matrix $H_i = \mathbb{E}[x_i x_i^\top]$ via its eigendecomposition:

- **Eigenvector** (u_k): Represents a specific "skill" or pattern of variation (e.g., a specific texture in images or topic in text).
- **Eigenvalue** (λ_k): Quantifies the **strength** or variance of that pattern in the dataset.

Spectral Assumption: "Shared Head, Disjoint Tail"

- **Shared Head** ($k \leq H$): Common eigenvectors $\{u_1, \dots, u_H\}$ with non-zero variance across all domains (universal knowledge).
- **Disjoint Tail** ($k > H$): Domain-specific eigenvectors $u_k^{(i)}$ with their eigenvalues following a **domain-specific power law**:

$$\lambda_k^{(i)} = \underbrace{h_i}_{\text{Weight}} \cdot \underbrace{k^{-\alpha_i}}_{\text{Power Law Decay}}$$

(Power-law decay is a standard assumption from prior work.)

The Loss Decomposition

Analyzing this one-pass SGD via continuous-time approximation and Stochastic Differential Equation (SDE):

Main Theorem:

The expected test loss on domain i can be decomposed as follows:

$$L_i(h, N, D) \approx \underbrace{c_i x_i^*(h)^{-b_i}}_{\text{Capacity Term}} + \underbrace{A_i (Dh_i)^{-a_i}}_{\text{Noise Term}} + E_i$$

1. Capacity Term

- Identical to Quantization Model.
- Driven by **competition** with other domains.
- Depends on **all** weights h .

2. Noise Term

- New term arising from stochastic gradient noise.
- Driven by **data volume**.
- Depends only on own weight h_i .

Explaining the Optimal Mixture

Why does the optimal training mixture h^* deviate from the target w ?

Mechanism: Noise Reduction

- **Harder Domains:** Domains with large A_i or small a_i contribute more variance to the loss.
- **Reweighting:** To minimize total loss, the optimization **shifts weights** h_i toward these "harder-to-learn" domains to suppress their variance.

Result 1: Superior Predictive Accuracy

Our theoretical models fit the observed loss landscape more accurately than existing empirical baselines, while using significantly fewer parameters.

Method	MRE (%)	MAE	# Param
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Extended Quantization	2.064	0.051	$3K$
Extended Linear Reg.	1.533	0.034	$5K$

Table 2: Fitting accuracy on 1B-parameter models ($K = 17$ domains).

Result II: Identifying the Optimal Mixture

We trained new models using the optimal mixtures predicted by each scaling law. Our method yielded the lowest validation loss on the target distribution.

Method	200M Model		700M Model	
	Val. Loss ↓	Perplexity ↓	Val. Loss ↓	Perplexity ↓
Uniform Baseline ($h_i = 1/K$)	2.228953	9.29	1.922253	6.836
Exponential Law	2.228489	9.285	-	-
BiMix (Ge et al., 2024)	2.229646	9.296	-	-
Additive Law (Shukor et al., 2025)	2.228093	9.282	1.922196	6.836
Ours (Eq. 4)	2.227662	9.278	1.921733	6.827

- **Efficiency:** Achieves these results using significantly fewer free parameters than empirical laws.

Future Work

Future Directions

1. Unseen Domains and Downstream Tasks

- **Current Limitation:** The model predicts loss only within the span of training domains.
- **Next Steps:** Extend the framework to predict how data mixtures affect performance on **unseen domains** and specific **downstream tasks**.

2. Explicit Domain Overlap Modeling

- **Current Limitation:** We assume "disjoint tails" for tractability.
- **Next Steps:** Develop models that explicitly quantify information overlap between highly correlated domains.

3. Reliable Fitting Algorithms

- **Current Limitation:** Fitting requires solving a non-convex bi-level optimization problem.
- **Next Steps:** Derive **convex relaxations** or **analytical approximations** to make parameter estimation faster and more robust to initialization.

Thanks and Questions?

Experimental Setup

We validate the Extended Linear Regression Model through two primary experiments:

Objective 1: Fitting Accuracy

- **Data:** 64 models (1B parameters) trained on 25B tokens.
- **Domains:** 17 domains from the Pile dataset.
- **Metric:** Mean Relative Error (MRE) on held-out mixtures.

Objective 2: Optimal Mixture Prediction

- **Data:** 200M and 700M parameter models.
- **Domains:** 4 domains (GitHub, PG-19, StackExchange, Wikipedia).
- **Goal:** Predict the mixture h^* that minimizes loss on a uniform target distribution.

Bi-Level Optimization Framework

We formulate the search for the optimal mixture h^* as a **bi-level problem**.

Inner Level: Implicit Capacity Allocation (Training)

Given a mixture h , the model allocates capacity $x^*(h)$ to minimize weighted training loss (ignoring noise):

$$x^*(h) = \arg \min_x \sum_{i=1}^K h_i c_i x_i^{-b_i} \quad \text{s.t.} \quad \sum x_i = N$$

Outer Level: Target Minimization (Validation)

We seek h^* to minimize loss on the **target distribution** w , which includes the critical **noise term** $A_i(Dh_i)^{-a_i}$:

$$h^* = \arg \min_{h \in \Delta^K} \sum_{i=1}^K w_i \left[\underbrace{c_i (x_i^*(h))^{-b_i}}_{\text{Capacity Term}} + \underbrace{A_i (Dh_i)^{-a_i}}_{\text{Noise Term}} \right]$$

Gradient Characterization

To optimize the outer objective $\mathcal{J}(h)$, we derive the gradient $\nabla_k \mathcal{J}$ via the Implicit Function Theorem.

The Gradient Decomposition:

$$\frac{\partial \mathcal{J}}{\partial h_k} = \underbrace{-w_k a_k A_k D^{-a_k} h_k^{-(a_k+1)}}_{\text{Force 1: Noise Reduction}} + \underbrace{\frac{\lambda x_k^*}{h_k(b_k + 1)} \left(\bar{R} - \frac{w_k}{h_k} \right)}_{\text{Force 2: Capacity Reallocation}} \quad (4)$$

- **Force 1 (Variance):** Directly encourages increasing h_k to reduce the noise term $A_k(Dh_k)^{-a_k}$ (requires more data).
- **Force 2 (Coupling):** Captures how changing h_k shifts the global shadow price λ , forcing a redistribution of capacity x^* across *all* domains.

We solve the non-convex outer optimization using **Online Mirror Descent** with exponentiated gradient updates.